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A dual-model deep learning method for sleep apnea detection based on representation learning and temporal dependence

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ABSTRACT

Sleep apnea (SA) is a sleep-breathing disorder accompanied by multiple complications. The SA detection method based on a single-lead electrocardiogram (ECG) has the characteristics of low power consumption and is desirable for the development of wearable equipment. This study proposed a dual-model deep learning method to perform representation learning and introduce long-term temporal dependence. First, the Christov algorithm was used to obtain the RR interval (RRI) of each 1-minute ECG segment, and the adaptive synthetic (ADASYN) sampling method was employed to synthesize the RRI series of the minority class to address the imbalanced learning problem. Then, a representation learning model based on the one-dimensional convolutional neural network (1DCNN-RLM) was built to extract the feature vector of the RRI series. Eventually, a temporal dependence model based on the bidirectional gated recurrent unit (BiGRU-TDM) was constructed to learn the state (SA/normal) transition pattern between the segments and complete the classification task. We employed the apnea-ECG database for experiments. For per-segment detection results, the accuracy, sensitivity, and specificity of this method were 91.1%, 88.9%, and 92.4%, respectively. The per-recording detection accuracy reached 100%. ADASYN alleviates the imbalance of sensitivity and specificity in classification results. The 1DCNN-RLM with powerful representation learning ability has extracted discriminative features. The BiGRU-TDM introduces the long-term time dependence of SA and improves classification performance. The results of this study substantiate that the proposed method is robust and has good transferability. This method provides a reference for the diagnosis of other diseases.

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1. Introduction

Sleep apnea (SA) is a common sleep-related disease primarily manifested by the collapse of the upper airway, which could lead to the repetitive drastic reduction or complete interruption of respiratory airflow. According to the American Academy of Sleep Medicine (AASM) standard: subjects with an apnea-hypopnea index (AHI) greater than or equal to 5 are considered SA patients [1]. Nearly 90% of SA patients do not receive timely diagnosis and treatment [2], and people with obesity and overweight are more likely to suffer SA [3]. Studies have confirmed that SA is normally accompanied by autonomic dysfunction [4], and it is likewise easy to cause complications such as hypertension [5], coronary heart disease [6], and heart failure [7].

Polysomnography (PSG) is internationally recognized as an effective instrument for SA detection. Subjects are required to wear up to 16 channels of electrodes for signal collection throughout the night, and then rely on experts for manual classification. This method is not only expensive but may also affect the sleep quality of subjects, which could further have an effect on result. Methods based on single-lead signals have attracted wide attention, such as blood oxygen saturation [8], photoplethysmography [9], snoring [10], respiration signals [11], electrocardiogram (ECG) [12], and so on. Compared with other signals, ECGs are convenient to collect and rich in breathing information. Studies reported that SA events can be reflected through ECGs [13–15]. There is a high correlation between heart rate variability and SA [16,17]. With the occurrence of SA events, the RR interval (RRI) obtained from ECGs will have certain periodic changes [18].

Most of the early methods based on ECGs concentrated on feature engineering, that is, the design of artificial features and the choice of different classifiers. Babaeizadeh et al. extracted spectral

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power features of instantaneous heart rate and used a quadratic classifier for SA segment detection [19]. Sadr et al. used the time domain and frequency domain features of the ECG-derived respiration and RRI, and the classifier was an extreme learning machine [20]. Varon et al. extracted the principal components of the QRS complexes and used the respiratory information, and then a least-squares support vector machine classifier was applied [21]. Song et al. took advantage of the temporal dependence of SA and designed a classifier combining support vector machines with a hidden Markov model [22]. The linear support vector machine classification methods based on artificial features have also attracted widespread attention [23,24]. Feature engineering highly relies on priori knowledge and usually requires experienced experts to design features. In short, traditional feature engineering methods require human intervention. Deep learning methods can avoid these restrictions as much as possible and use the network structure and learning strategy to make up for the incompleteness of artificial features [25].

Deep neural networks, such as the convolutional neural network (CNN) and recurrent neural network (RNN), have enabled many researchers to achieve great success in the fields of image classification [26], image reconstruction [27], action recognition [28], language modeling [29] and natural language processing (NLP) [30]. In recent years, many researchers have begun to use deep learning methods to establish SA classification models. Some researchers suggested the use of stacked autoencoders to extract the features of RRI and proposed their algorithm-level solutions for the imbalanced learning problem [31,32]. Singh et al. performed the continuous wavelet transform on each 1-minute ECG segment to generate scalogram images and then proposed a classification model based on a two-dimensional CNN (2DCNN) [33]. The one-dimensional CNNs (1DCNN) have also attracted the attention of researchers [34,35]. Studies [26–30] have shown that CNNs have good representation learning ability and the RNNs play an exceedingly important role in temporal sequence learning. In other works, Erdenebayar et al. used different types of neural network for modeling and then recommended the 1DCNN and gated recurrent unit (GRU) for automatic SA detection [36]. However, the GRU is only used for obtaining the temporal information inside the segment [36], which does not seem to make full use of the GRU's modeling ability of the temporal sequence. For multi-modal sleep scoring, Zhang et al. [37] combined traditional RNNs and Markov chain model to introduce the temporal correlations between consecutive epochs. However, traditional RNN has gradient disappearance and gradient explosion problems [38]. Markov chain model require initial distribution and transition matrix to initialize. GRU neither has the problem of gradient disappearance and gradient explosion, nor does it rely on any priori probability. Based on previous results, this study proposes a new idea to consider the respective advantages of the CNNs and GRUs for network model combination.

The significance of this study mainly has three points:

1) Researchers' solutions to imbalanced learning problem are generally based on classifiers [31,32]. These methods seem to sacrifice classification accuracy and increase the complexity of the models. We adopt the data-level adaptive synthetic (ADASYN) sampling method [39] to deal with this problem and alleviate the imbalance of sensitivity and specificity in classification results.
2) For the purpose of reducing the reliance on priori knowledge in the feature extraction process, we construct a 1DCNN-based representation learning model (1DCNN-RLM) to automatically extract the discriminative feature vector of the RRI series. In the 1DCNN-RLM, we use the improved 1D-inception blocks for multi-scale feature fusion, which ultimately improves the representation learning ability of this model.
3) To introduce the temporal dependence of SA, researchers increased the segment length [4,40,41] or use the Markov model [22,31,42]. The temporal information introduced by the former is very limited. The latter needs to obtain the priori probability of the state transition. Therefore, we propose a bidirectional GRU-based temporal dependence model (BiGRU-TDM) to introduce long-term temporal dependence of SA and improve classification performance. The improved Self-attention layers in the BiGRU-TDM can highlight key information between the time steps and inside each time step to some extent.

The remainder of the paper is structured as follows. Section 2 mainly presents the details of the ADASYN, 1DCNN-RLM and BiGRU-TDM. Section 3 presents a comprehensive and fair analysis on the testing, validation, and generalization of the proposed method. Section 4 discusses the results, and we conclude this study in Section 5.

2. Materials and methods

2.1. Complete framework of the proposed method

The schematic diagram of the proposed method is shown in Fig. 1. The beginning step is to preprocess the single-lead ECG signals. After obtaining the RRI series of each 1-minute ECG segment, the train set is used to train the 1DCNN-RLM and BiGRU-TDM. ADASYN is applied during 1DCNN-RLM training. The feature vectors extracted by the trained 1DCNN-RLM are used for BiGRU-TDM training. In regard to the test set, per-segment detection is performed using the trained models. Then, the predicted label is used to complete the AHI calculation to perform the per-recording detection.

2.2. Data and preprocessing

In view of the characteristics of heart rate variability with its own variations and large individual differences [43], this study considers the apnea-ECG database provided by PhysioNet [44,45] to conduct a comparative analysis under the same conditions as much as possible. To verify the transferability of the proposed method, we collected another database named the In-group database from The Sixth Affiliated Hospital of Sun Yat-sen University for the generalization experiment.

The description of the apnea-ECG database and In-group database is presented in Table 1. The apnea-ECG database is a public database, including 70 overnight single-lead ECG recordings (35 in the released set and 35 in the withheld set). The duration of these ECG recordings is between 401 and 578 min and the sampling rate is 100 Hz. Experts annotated whether SA event occurred in 1-minute ECG segments. The entire database contains a total of 34,428 min of data, but only 34,230 min of data are annotated (0: normal, 1: SA). All results except those from the generalization experiment are based on the apnea-ECG database. The In-group database includes 105 overnight single-lead ECG recordings. The duration of the ECG recordings is between 360 and 587 min, and the sampling rate is 500 Hz. Hospital experts combined with other signals (including respiration and oxygen saturation) in PSG to annotate whether SA event occurred in each ECG segment. The In-group database is used for the generalization experiment. In the generalization experiment, all 105 recordings in the In-group database are used as the train set and all 70 recordings in the apnea-ECG database are used as the test set.

The detailed description of the data preprocessing steps is as follows: 1) Correct the ECG recordings of the reversed positive

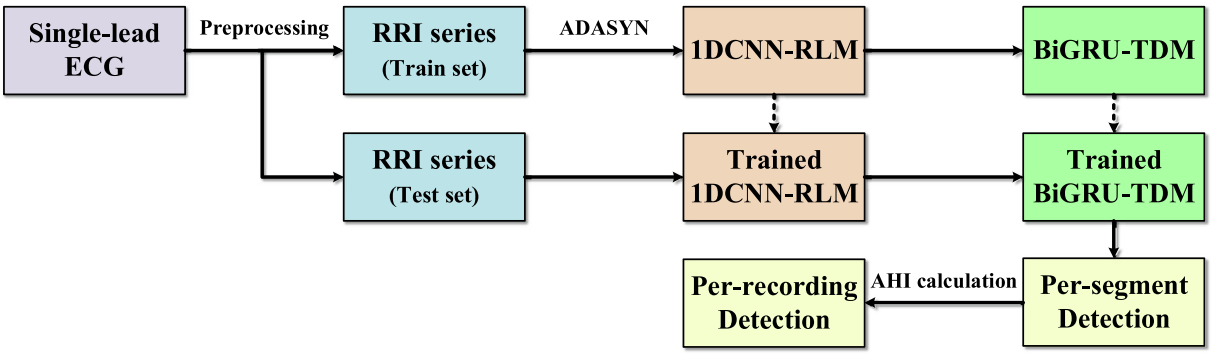


Fig. 1. Schematic diagram of the proposed method. 1DCNN-RLM: representation learning model based on the one-dimensional convolutional neural network; BiGRU-TDM: temporal dependence model based on the bidirectional gated recurrent unit; ADASYN: adaptive synthetic sampling method.

and negative electrodes. 2) Separate the ECG recordings into 1-minute segments corresponding to the label. 3) Eliminate ECG segments missing more than 20 s. 4) Use the Christov algorithm [46] (it includes noise suppression and can adapt to different sampling rates) to detect R waves to obtain the RRI series from each 1-minute ECG segment. 5) Eliminate abnormal RRI series with lengths other than 30–150. 6) Use the median filter proposed by Chen et al. [47] to eliminate the RRI singular values outside [0.4, 2] in each series. 7) Use cubic spline interpolation for all RRI series to unify to 256 points. The final released set retains 16,875 RRI series (SA: 6,505, normal: 10,370), the withheld set retains 17,117 RRI series (SA: 6,518, normal: 10,599). The In-group database retains 52,004 RRI series (SA: 15,637, normal: 36,367).

After data preprocessing, the corresponding AHI values of each ECG recording are calculated according to the finally obtained label information. For each ECG recording, the AHI calculation formula is as follows [31,32].

$$\mathrm{AHI} = \frac{60N_{\mathrm{SA}}}{T} \tag{1}$$

where T represents the number of labels, and N_{SA} represents the number of SA segments with a label of 1.

According to the AHI standard of the AASM, 23 recordings in the released set and withheld set both have an AHI $\geq$ 5, and the remaining 12 recordings have an AHI < 5. There are 74 recordings in the In-group database with an AHI $\geq$ 5, and the remaining 31 recordings with an AHI < 5.

2.3. Representation learning model

In the domain of biomedical engineering, the traditional 2DCNN is widely applied in medical images [48]; in contrast, the 1DCNN is mainly used in physiological signals [49]. The 1DCNN takes one-dimensional data as input. The convolutional layer and pooling layer have a one-dimensional form. In this study, the one-dimensional convolution (Conv1D) layer and one-dimensional max pooling (MaxPool1D) layer are used to construct the representation learning model. In addition, the parametric rectified linear unit (PReLU) [50] is used as the activation function of Conv1D layer.

The RRI series shows a certain difference between the SA events and normal sleep [31]. This difference can be used to obtain the discriminative features between SA segments and normal segments. As shown in Fig. 2, we constructed the 1DCNN-RLM to complete representation learning and used the trained model to extract the feature vectors of the RRI series. The Inception block has made great progress and achieved great success in the field of computer vision [51–53]. The Inception block can realize the fusion of multi-scale features while increasing the network width. In addition, the block uses a convolutional layer with a kernel size of 1 to reduce the amount of calculation. Considering that the research object is a one-dimensional signal, we transform the original Inception block into a one-dimensional form. Then, an improved 1D-inception block is designed based on the original block, as shown in Fig. 3. The improved 1D-inception block can complete deeper feature fusion. The block has 4 branches, and each branch is composed of the Conv1D layer or the MaxPool1D layer.

1DCNN-RLM takes the 256 $\times$ 1 RRI series as the model input. To simplify the network complexity, the model applies the MaxPool1D layer in three locations. Before each improved 1D-inception block, a Dropout operation with a dropout rate of 0.5 is added, which is usually applied to improve the generalization ability of the network [54]. The role of the Flatten operation is to adjust the features of 256 channels into a one-dimensional feature vector. The Dense layer (usually called the fully connected layer) adds L2 normalization (θ = 0.0001) to reduce the risk of overfitting. Last, there is a Softmax layer, which can directly output the predicted probability. After the 1DCNN-RLM completes the training, we apply the parameters from the model input to the Flatten operation to extract the feature vectors (dimension: 1024). Then, the feature vectors will be used as the model input of the BiGRU-TDM.

Ordinarily, the ratio of SA samples to normal samples is imbalanced. Then, models will tend to learn from majority class, which may result in providing unfavorable accuracies across the data classes [55]. ADASYN, as a commonly used sample synthesis algorithm, can be used to deal with the imbalanced learning problem [39]. We use ADASYN to synthesize the RRI series belonging to the minority class in the train set and then use the synthetic RRI series and the original RRI series together as the input of the 1DCNN-RLM for training (including the parameter optimization stage and the model training stage). In this study, ADASYN is applied to synthesize enough samples to keep the absolute balance of the sample sizes of two classes. The ADASYN implementation steps can be found in [39].

2.4. Temporal dependence model

The traditional RNN cannot learn long-term dependence due to gradient disappearance and gradient explosion problems [38], so Cho et al. designed the GRU to solve these problems [56]. The

Table 1
The information of the databases.

Database	Recording number	Length (minute)	Annotation number
apnea-ECG	70	491.8 ± 31.6	34,230
In-group	105	498.1 ± 45.2	52,302

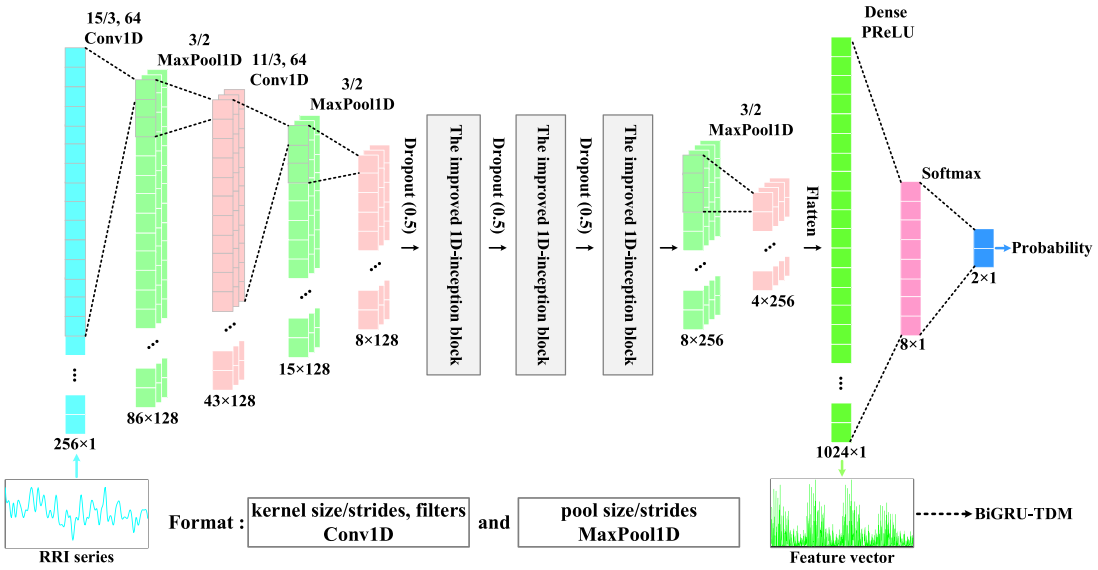


Fig. 2. The structure of the 1DCNN-RLM. The 1DCNN-RLM is mainly composed of 2 Conv1D layers, 3 MaxPool1D layers, and 3 improved 1D-inception blocks. During the training process, the label information of the train set is used. The trained model is mainly used to extract feature vectors from the RRI series.

GRU has the excellent modeling ability of temporal sequences. Therefore, GRUs are suitable instruments for introducing long-term time dependence. The internal structure of the GRU is shown in Fig. 4. In theory, BiGRU can obtain more information than the single-directional GRU. The BiGRU has an additional set of units that propagate in the opposite direction to that of the single-directional GRU [57]. This work uses BiGRU as the main component to build the temporal dependence model.

Recently, the self-attention mechanism has gained remarkable achievements in temporal sequence learning tasks such as translation [58] and semantic role labeling [59]. In related research on GRUs, the Self-attention layer based on the self-attention mechanism generally directly acts on the output of the GRU, thereby highlighting some important information in the time dimension. To enhance the attention to the internal information of each single time step, we designed an improved Self-attention layer, as shown in Fig. 5.

Assuming that the BiGRU has T time steps, and the t time step input of the improved Self-attention layer is $\boldsymbol{h}_t$, the calculation formula for the corresponding time step output $\boldsymbol{v}_t$ is as follows.

$$\boldsymbol{I}_t = softmax(\boldsymbol{W}_1 \cdot \boldsymbol{h}_t + b_1) \tag{2}$$

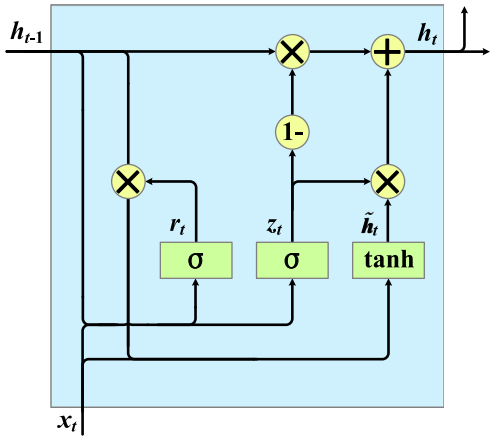


Fig. 4. The internal structure of the GRU. The GRU has two basic gates: reset gate and update gate. The reset gate is used to reduce the irrelevant information in the previous unit, and the update gate is used to determine how much information from the previous unit needs to be passed to the next unit.

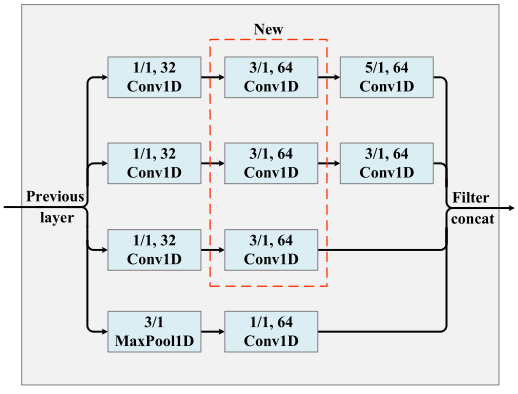


Fig. 3. The improved 1D-inception block. Compared with the original 1D-inception block, a Conv1D layer with a convolution scale of 3 and a channel number of 64 is added to the three branches other than the branch containing the MaxPool1D layer.

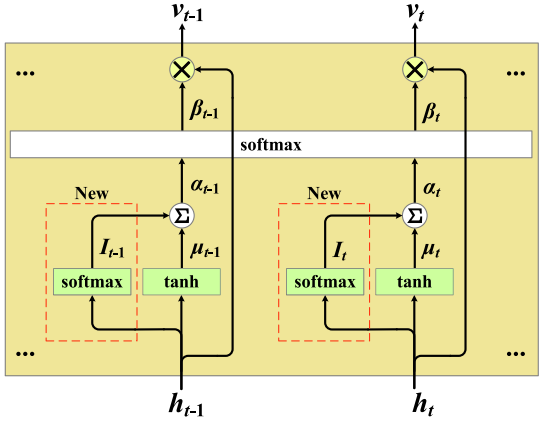


Fig. 5. The improved Self-attention layer. Different weights are set for the information on different dimensions within a single time step, and then the attention weights for that time step are formed.

$$\boldsymbol{\mu}_t = tanh(\boldsymbol{W}_2 \cdot \boldsymbol{h}_t + b_2) \tag{3}$$

$$\alpha_t = \boldsymbol{I}_t^{\mathrm{T}} \cdot \boldsymbol{\mu}_t \tag{4}$$

$$[\beta_1; \cdots; \beta_t; \cdots; \beta_T] = softmax([\alpha_1; \cdots; \alpha_t; \cdots; \alpha_T]) \tag{5}$$

$$\boldsymbol{v}_t = \beta_t \boldsymbol{h}_t \tag{6}$$

where the weight matrix $\boldsymbol{W}$ and the bias value b are shared at each time step. $\boldsymbol{I}_t$ represents the degree of attention in different dimensions within the same time step at t, and $\boldsymbol{\mu}_t$ represents the internal information of the time step. β_t stands for the attention degree of time step t.

SA segments in minutes have a greater potential to appear continuously over time rather than appear randomly scattered, which indicates that there may be a state (SA/normal) transition pattern between the segments [22]. Thus, this study constructs the BiGRU-TDM, which is used to learn the pattern. The structure of BiGRU-TDM is shown in Fig. 6. The vectors input to BiGRU-TDM can be obtained by moving a sliding window with a length of 25 and a span of 1. The order of the vectors is consistent with the original order of RRI series. The model also uses the Dropout operation and L2 normalization to reduce overfitting. The trained BiGRU-TDM can directly output the predicted probability to carry out per-segment detection.

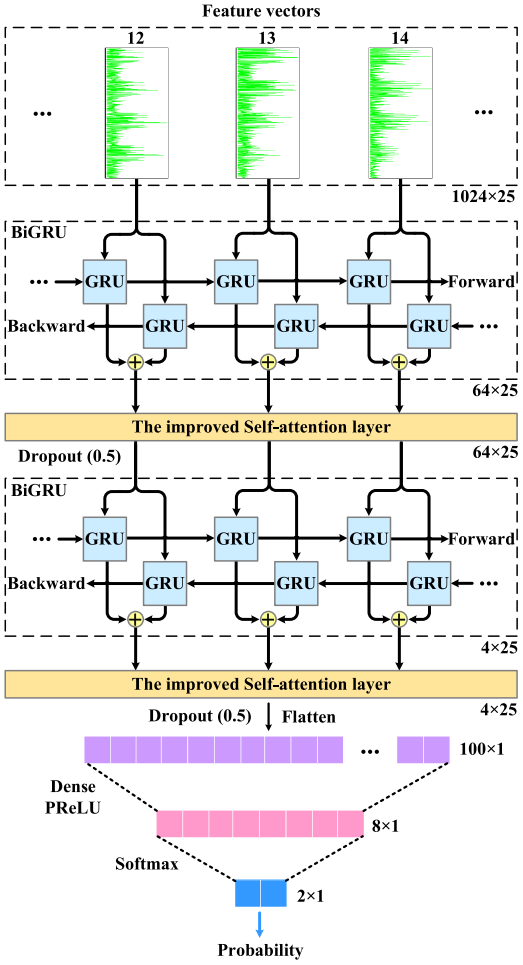


Fig. 6. The structure of the BiGRU-TDM. The BiGRU-TDM is mainly composed of two BiGRU layer with a time step of 25 (the outputs in the two directions are combined in an additive manner). The improved Self-attention layer is added after each BiGRU layer.

In this work, 1DCNN-RLM extracts features segment by segment (1-minute). However, in order to introduce long-term temporal dependence, BiGRU-TDM has a time step length of 25 (corresponding to consecutive 25-minute segments). It is worth noting that the label used in BiGRU-TDM training is the label of the thirteenth feature vector, so that the model can capture up to 12 min of temporal information before and after the segment. This is why we did not adopt an end2end framework. If 1DCNN-RLM and BiGRU-TDM are jointly trained, then for a certain minute segment, BiGRU-TDM may be able to introduce the previous temporal information through the memory but cannot obtain the subsequent temporal information. Therefore, our study proposes this dual-model method, which is based on the consideration of BiGRU-TDM function realization.

2.5. Performance evaluation

To comprehensively evaluate the performance of the proposed method in this study, we used the following performance indexes.

1) Accuracy

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{7}$$

where TP, TN, FP, and FN represent "true positive", "true negative", "false positive", and "false negative", respectively.

2) Sensitivity (also known as recall).

$$\text{sensitivity} = \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

3) Specificity.

$$\text{specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{9}$$

4) Precision.

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{10}$$

5) F1 score.

$$\text{F}_1 = \frac{2\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{11}$$

6) Precision-recall (PR) curve.
7) Receiver operating characteristic (ROC) curve.
8) Area under curve (AUC).
9) Mean absolute error (MAE), which is introduced by Feng et al. [32].

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} |\text{trueAHI}_i - \text{predictedAHI}_i| \tag{12}$$

where N represents the number of ECG recordings in the test set.

Table 2
The hyperparameters of the models.

Hyperparameter	1DCNN-RLM	BiGRU-TDM
epoch	128	64
batch size	512	128
learning rate	0.0005	0.00015
loss function	cross-entropy	
optimizer	Adam	
beta1	0.9	
beta2	0.999	
epsilon	1e-8	

beta1: the exponential decay hyperparameter for the first moment estimates in the Adam optimizer; beta2: the exponential decay hyperparameter for the second moment estimates in the Adam optimizer; epsilon: the hyperparameter preventing division by zero in Adam updates.

10) The correlation coefficient between the true AHI and the predicted AHI.

3. Results

3.1. Hyperparameters

The hyperparameter optimization, training, and testing of the models are implemented by utilizing Python3.6 with TensorFlow1.10 deep learning library. As demonstrated in Table 2, some of the training parameters in 1DCNN-RLM and BiGRU-TDM are the same. And two models use the mini-batch gradient descent strategy. In the 1DCNN-RLM, the number of training epochs is set to 128, the batch size is 512, the initial learning rate is 0.0005, and the learning rate schedule is used (decreasing the learning rate by 10% every 8 epochs). In BiGRU-TDM, the number of training epochs is set to 64, the batch size is 128, the initial learning rate is 0.00015, and the learning rate schedule is also used (decreasing the learning rate by 2% every 16 epochs). Both models are trained using the Adam optimizer (beta1 = 0.9, beta2 = 0.999, epsilon = 1e-8) and use cross-entropy as the loss function. Based on 35 single-lead ECG recordings of the released set, we use the leave-one-record-out (LORO) cross-validation method to optimize the hyperparameters and determine the model structure (during this process, the 1DCNN-RLM used the ADASYN sampling method).

3.2. Balanced learning and model combination

After using the released set of the apnea-ECG database for model training, the test results of the withheld set are shown in Table 3. Scheme 1 is the per-segment detection result of 1DCNN-RLM when ADASYN is not used to synthesize the RRI series of the minority class. Scheme 2 is the per-segment detection result of 1DCNN-RLM when ADASYN is employed. Scheme 3 and Scheme 4 are the per-segment detection results combined with BiGRU-TDM based on Scheme 1 and Scheme 2, respectively. Using ADASYN does not contribute to an obvious increase in accuracy, but the sensitivity and specificity of Scheme 2 are more balanced than that of Scheme 1. After the introduction of long-term temporal dependence, the indexes of Scheme 4 relative to Scheme 2 have been significantly improved. It is worth noting that the balance of Scheme 4 is also better than that of Scheme 3. The PR curves of the above four schemes are shown in Fig. 7(a). The PR curve of Scheme 4 is almost above the other three PR curves and has the highest AUC (PR). These results show the effectiveness of the ADASYN and network model combination. BiGRU-TDM has a great effect on detection performance.

3.3. Overall test results on the withheld set

To facilitate peer comparison, we followed the rules of Computers in Cardiology Challenge in 2000, that is, using the released set of the apnea-ECG database to train the model and then conducting a test experiment based on the withheld set. This is also the main result demonstrated by the proposed method. The results of per-segment detection and per-recording detection are shown in Table 4. The segment classification accuracy reached 91.1%, and the per-recording detection accuracy achieved 100%. As shown in Fig. 7(b), we have drawn the ROC curve of the per-segment detection results to show the overall performance of the model.

3.4. Tenfold cross-validation

In this study, we also used 70 recordings of the apnea-ECG database to perform tenfold cross-validation for per-segment detection. Moreover, 70 ECG recordings were divided randomly into 10 sets in which 9 of them were used as training data and 1 was used as test data in turn for experimentation. Tenfold cross-validation can prove to a certain extent whether the model is overfitting.

The boxplots of the results are shown in Fig. 8. The average value of each index is very close to the per-segment detection results in Table 4, indicating that the proposed method has a certain degree of robustness.

3.5. Generalization experiment

After finishing the training, testing, and validating based on the apnea-ECG database, in order to further prove the generalization capability of the proposed method, we use all 105 recordings in the In-group database to train the model and then complete the test work based on 70 recordings in the apnea-ECG database. Generalization experiment between two sets of data from different sources can strongly prove the transferability of the method.

The results of per-segment detection and per-recording detection are shown in Table 5. The per-segment detection accuracy is 88.7%. For per-recording results, only an SA patient was misclassified. The generalization experiment results based on two different databases confirm that the proposed method has good transferability.

4. Discussion

4.1. Comparison with previous work

We compare the proposed method with 11 other methods, of which 6 are feature engineering methods, and the rest are deep

Table 3
Per-segment detection results of schemes without and with the ADASYN on the withheld set. Scheme 4 is the proposed method.

Scheme	Acc. (%)	Sen. (%)	Spe. (%)	Pre. (%)	F_1	AUC (PR)
1. 1DCNN-RLM without the ADASYN	83.8	76.5	88.2	80.0	0.782	0.863
2. 1DCNN-RLM with the ADASYN	84.2	81.7	85.7	77.9	0.798	0.868
3. 1DCNN-RLM without the ADASYN + BiGRU-TDM	89.6	81.2	**94.8**	**90.6**	0.856	0.936
4. 1DCNN-RLM with the ADASYN + BiGRU-TDM	**91.1**	**88.9**	92.4	87.8	**0.883**	**0.952**

Acc.: accuracy; Sen.: sensitivity; Spe.: specificity; Pre.: precision; F_1: F1 score; AUC (PR): area under the precision-recall curve.

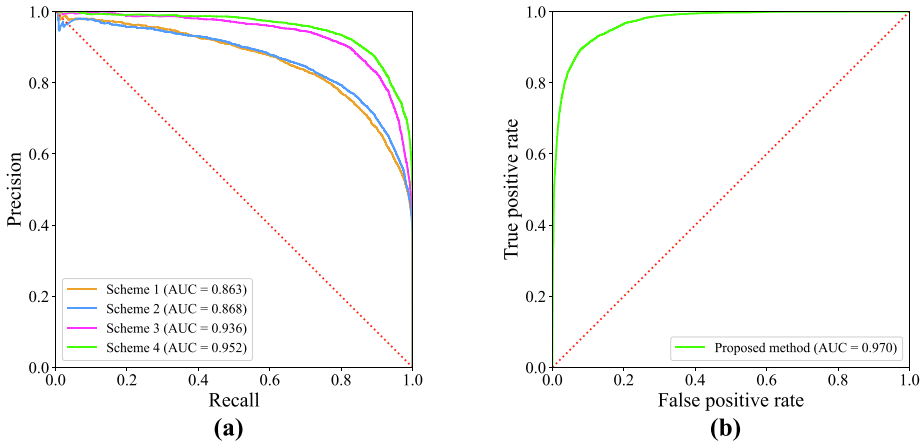


Fig. 7. (a) Precision-recall curves of schemes without and with the ADASYN on the withheld set. (b) Receiver operating characteristic curve of the proposed method on the withheld set.

Table 4
Per-segment and per-recording detection results of the proposed method on the withheld set.

Per-segment detection						Per-recording detection						
Acc. (%)	Sen. (%)	Spe. (%)	Pre. (%)	F_1	AUC (ROC)	Acc. (%)	Sen. (%)	Spe. (%)	Pre. (%)	F_1	MAE	Corr.
91.1	88.9	92.4	87.8	0.883	0.970	100	100	100	100	1.000	3.333	0.968

AUC (ROC): area under the receiver operating characteristic curve; MAE: mean absolute error; Corr.: correlation coefficient.

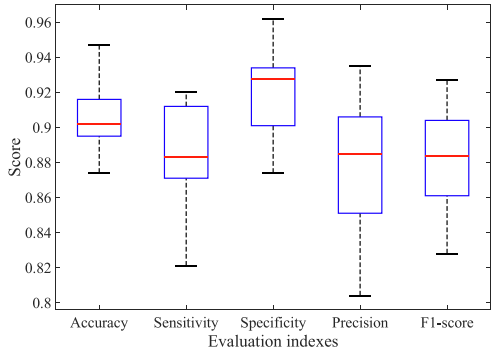


Fig. 8. The boxplots of per-segment detection evaluation indexes in tenfold cross-validation. Accuracy is the most stable index, and all indexes have no outliers.

learning methods. These methods all use the released set for model training and the withheld set for model testing. The comparison of per-segment detection results is presented in Table 6. Under the same conditions, the classification accuracy, specificity and AUC (ROC) of the proposed method is better than that of the others, and the sensitivity is also higher than the results of most studies. It is worth noting that the classification accuracy is very close to the highest accuracy of manual detection (92.6%) [60].

Based on the per-segment detection results, the AHI of each recording is calculated by Eq. (1). The predicted AHI is compared with the true AHI to obtain the per-recording detection results. The results are shown in Table 7. Sadr et al. [20] did not provide their per-recording detection results. Some studies did not use the AASM standard to classify recordings. In addition, Babaeizadeh [19] did not use all 35 recordings in the withheld set for model testing. Under the AASM standard, the proposed method can completely distinguish 23 SA recordings and 12 normal recordings in the withheld set. The MAE and correlation coefficient between the predicted AHI and true AHI are better than other methods.

In order to follow the rules of Computers in Cardiology Challenge in 2000 and facilitate peer comparison, the ECG recordings in this work is segmented by a 1-minute window the same as other work in Table 6. This study did not consider the window length as a potential influencing factor, but this issue deserves more attention. Some researchers have begun to use different window lengths to segment the ECG recordings. Based on the apnea-ECG database, Li et al. [4] divided the data into 5-minute segments and then extracted the sliding trend fuzzy approximate entropy for per-recording detection. Pietrzak et al. [40] applied a window length of 10,000 s, and then performed the per-recording detection based on standard deviation of successive differences. Hayano et al. [41] used a window length of 60-minute and then analyzed the cyclic variation of heart rate of the segments. These studies all did not provide the per-segment and per-recording detection results of the withheld set according to the challenge agreement, so our work did not compare with them. For the optimization of the window length, we believe that more comprehensive and systematic experiments and discussions are required. In the future SA research, we may take the window length into consideration.

4.2. Imbalanced learning problem

In binary classification tasks, the imbalance of sensitivity and specificity caused by sample imbalance is very common. This

Table 5
Per-segment and per-recording detection results of the generalization experiment.

Per-segment detection						Per-recording detection						
Acc. (%)	Sen. (%)	Spe. (%)	Pre. (%)	F_1	AUC (ROC)	Acc. (%)	Sen. (%)	Spe. (%)	Pre. (%)	F_1	MAE	Corr.
88.7	86.2	90.3	84.7	0.854	0.951	98.6	97.8	100	100	0.989	4.762	0.934

Table 6
Comparison of per-segment detection results between the proposed method and other methods on the withheld set.

Type	Reference	Acc. (%)	Sen. (%)	Spe. (%)	AUC (ROC)
FE	Babaeizadeh et al. [19] (2010)	84.7	76.7	89.6	–
	Sadr et al. [20] (2014)	87.7	81.3	91.7	–
	Varon et al. [21] (2015)	84.7	84.7	84.7	0.881
	Song et al. [22] (2015)	86.2	82.6	88.4	0.940
	Sharma et al. [23] (2016)	83.8	79.5	88.4	0.834
	Surrel et al. [24] (2018)	82.2	73.3	87.6	–
DL	Li et al. [31] (2018)	84.7	88.9	82.1	0.869
	Wang et al. [35] (2019)	87.6	83.1	90.3	0.950
	Singh et al. [33] (2019)	86.2	**90.0**	83.8	0.881
	Feng et al. [32] (2020)	85.1	86.2	84.4	0.860
	Shen et al. [42] (2021)	89.4	89.8	89.1	0.964
	Proposed method	**91.1**	88.9	**92.4**	**0.970**

FE: feature engineering; DL: deep learning.

occurs because models or algorithms fail to properly represent the distributive characteristics of the data [55]. In this study, we use ADASYN to deal with the imbalanced learning problem. When optimizing and training the 1DCNN-RLM, the number of SA RRI series is equal to that of normal RRI series, in which case the imbalanced learning problem could be alleviated to some degree. It can be observed from Table 3 that the gap between sensitivity and specificity of 1DCNN-RLM is significantly reduced after using ADASYN. It is worth noting that regardless of whether ADASYN is used, the accuracy of 1DCNN-RLM has not changed obviously. The reason is that, as a sampling method, ADASYN cannot contribute additional useful information. Nevertheless, the value of ADASYN can be seen in the final classification results. The classification accuracy of Scheme 4 is improved compared with Scheme 3, and it has higher sensitivity. As shown in Fig. 7(a), although Scheme 4 has an intersection with the other PR curves, the AUC (PR) value of Scheme 4 is higher than the other three schemes. The above results prove that ADASYN can alleviate the imbalance between sensitivity and specificity in the classification results of 1DCNN-RLM and additionally causes a certain effect on the final classification results.

4.3. Representation learning for feature extraction

In feature engineering, the process of designing and selecting artificial features need the direct participation of experts. In addition, it is necessary to select a suitable classifier. These all require laborious effort and priori knowledge. Applying deep learning methods to feature extraction has certain advantages over feature engineering methods. High-dimensional and complex features can be extracted easily by the neural network, while difficultly by human experts [25]. Moreover, the deep learning methods can reduce the human intervention in the preliminary work, making the entire process more automated.

Studies have shown that the CNN is an effective tool for feature extraction [26], and the 1DCNN is a type of convolutional network for one-dimensional signals. A previous study found that the low-frequency component of the RRI series will increase when an SA event occurs. This low-frequency component is caused by bradycardia during sleep apnea followed by tachycardia upon its cessation [18]. Therefore, we constructed a 1DCNN-based representation learning model for feature extraction based on this characteristic. The RRI series are directly used for BiGRU-TDM training, but it does not perform well. The possible cause is that the morphology of the RRI series is too diversified and the information is redundant, which ultimately leads to the failure of the network to achieve convergence. This phenomenon supports the importance of representation learning from the side [61]. The feature vectors obtained by representation learning have a higher degree of discrimination, which provides convenience for BiGRU-TDM to introduce the temporal dependence of SA.

4.4. Temporal dependence of sleep apnea

Researchers usually use the Markov model to introduce the time dependence of SA [22,31,42]. There are also studies that just increase the length of the ECG segments [4,40,41]. However, the former needs to obtain a priori assumption about the data distribution. The temporal information introduced by the latter is very limited. To dispense with these restrictions as much as possible, we suggest using deep learning methods to introduce long-term temporal dependence of SA. As shown in Tables 6 and 7, the per-segment and per-recording detection results of the proposed method are better than those of the method using hidden Markov model [22,31,42]. This shows that the temporal dependence model based on BiGRU and self-attention mechanism can replace the Markov model to introduce long-term temporal dependence and achieve better results.

We have built models based on bidirectional long short-term memory (BiLSTM) and BiGRU under the same conditions, and found that their test results are very close. Considering that the network parameters of BiLSTM are about 30% more than BiGRU, we finally chose BiGRU. After optimizing the structure of BiGRU-TDM, the time steps of BiGRU are determined to be 25. With the introduction of the self-attention mechanism, we think that the step length is self-adaptive within the fixed time steps because the weight of a time step can be trained to 0. Table 3 shows the promotion effect of BiGRU-TDM on per-segment detection performance. The MAE and correlation coefficient of the proposed method is better than the other three methods. Fig. 9(a) shows 25-minute consecutive output probabilities in the withheld set (the first 10 min are normal, and the SA event occurs continuously

Table 7
Comparison of per-recording detection results between the proposed method and other methods on the withheld set.

Reference	Acc. (%)	Sen. (%)	Spe. (%)	MAE	Corr.	AASM standard	Recording number
Babaeizadeh et al. [19] (2010)	100	100	100	–	–	No	30
Varon et al. [21] (2015)	100	100	100	–	–	No	35
Song et al. [22] (2015)	97.1	95.8	100	–	0.860	Yes	35
Sharma et al. [23] (2016)	97.1	95.8	100	–	0.841	Yes	35
Surrel et al. [24] (2018)	91.3	92.7	89.2	–	–	No	35
Li et al. [31] (2018)	100	100	100	–	–	Yes	35
Wang et al. [35] (2019)	97.1	100	91.7	–	0.943	Yes	35
Singh et al. [33] (2019)	100	100	100	–	–	Yes	35
Feng et al. [32] (2020)	97.1	95.7	100	5.600	–	Yes	35
Shen et al. [42] (2021)	100	100	100	4.230	–	Yes	35
Proposed method	**100**	**100**	**100**	**3.333**	**0.968**	**Yes**	**35**

AASM standard: AHI greater than or equal to 5 is deemed to be a patient; Recording number: the number of the withheld set ECG recordings used in testing their method.

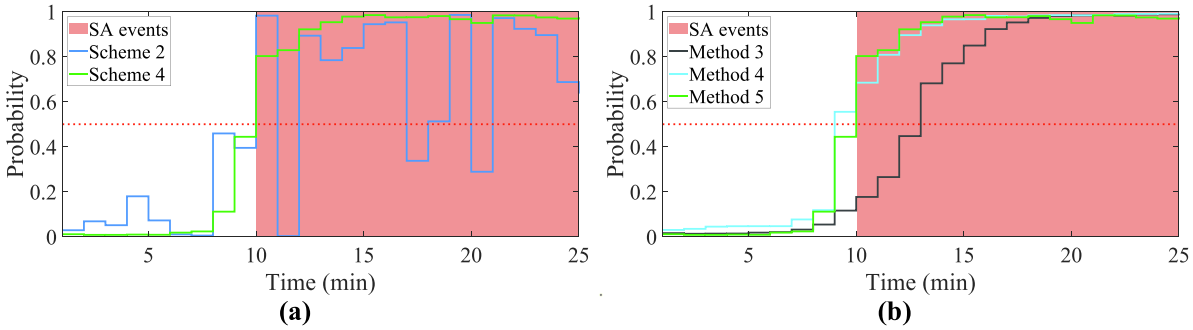


Fig. 9. (a) 25-minute consecutive output probabilities of different schemes in the withheld set. (b) 25-minute consecutive output probabilities of different methods in the withheld set. The proposed method in the figure is equivalent to Scheme 4 and Method 5. Probability represents the predicted probability of an SA event occurring in the segment.

in the next 15 min). It can be observed that the output probabilities of the proposed method are better than that of Scheme 2 (refer to Table 3). The proposed method greatly improves the accuracy of segment classification and maintains the probabilities close to the label. This result directly shows the effect of introducing long-term temporal dependence of SA.

Zhang et al. [37] used RNN to introduce temporal correlation and Markov chain to complete probability inference. Traditional RNN has the problem of gradient disappearance and gradient explosion [38], so it is not suitable for introducing long-term temporal dependence. The limitation of the Markov chain model lies in the need to obtain the initial distribution and transition matrix to initialize. Improved RNNs such as GRUs has overcome the problem of gradient [56] and does not depend on any priori probability. As shown in Table 8, on the basis of 1DCNN-RLM, we compared the results of the per-segment detection that use different components to introduce long-term temporal dependence. The proposed method has the best comprehensive performance. Both the traditional RNN and Markov chain can improve the detection results, but the combination of the two only brings a slight improvement. It is worth noting that the use of Markov chain after BiGRU-TDM can no longer bring significant performance improvements. Therefore, we have reason to believe that BiGRU-TDM can replace the traditional RNN and Markov chain to achieve better results, at least in the field of SA detection.

4.5. Motivation for the dual-model strategy

There are two considerations for adopting the dual-model strategy in this paper.

1) Convolutional neural networks are good at feature extraction. The gated recurrent unit has excellent modeling ability of the temporal sequence. The focus of the two is different. In this paper, 1DCNN-RLM based on convolutional neural network is constructed for multi-scale feature fusion and extraction, and BiGRU-TDM based on gated recurrent unit is constructed to introduce long-term temporal dependence between segments. We hope that the two models can make full use of their respective advantages to achieve better expectations. In addition, the joint training of two different types of networks may cause mutual constraints to some extent. Therefore, this article adopts a dual-model strategy rather than an end-to-end strategy.
2) Limited by the training mechanism of the neural network, the proposed method cannot adopt an end-to-end strategy. During the training process, 1DCNN-RLM extracts the feature vector of each RRI sample. The input of BiGRU-TDM is 25 consecutive feature vectors, and the training target is the label of the thirteenth feature vector. BiGRU-TDM can introduce the twelve-minute temporal dependence information before and after a segment. If an end-to-end strategy is adopted, 1DCNN-RLM and BiGRU-TDM are trained simultaneously. For a certain RRI sample, although BiGRU-TDM can import the past feature vectors as input through the history cache, the feature vectors after the sample cannot be imported because 1DCNN-RLM has not yet performed feature extraction. In order to achieve the desired function, we adopted a dual-model strategy.

Zhang et al. used 1DCNN and 2DCNN in the dual-CNN model to extract the features of time domain and time–frequency domain, and then used a combination of RNN and Markov chain to introduce temporal dependence [37]. Zhang et al. constructed a "dual-CNN" model to extract features to avoid reliance on priori knowledge, but adopted Markov chain that requires priori probability in the second phase. This seems contradictory. In terms of the basic idea of "dual-model", the motivation in Zhang's article is similar to that in this article. However, there are obvious differences between this article and Zhang's article on the scheme of introducing temporal dependence, which has been discussed in Section 4.4. In a sense, BiGRU can replace traditional RNN and Markov chain to introduce long-term time dependence.

Table 8
Comparison of per-segment detection results that use different components to introduce long-term temporal dependence.

Method	Acc. (%)	Sen. (%)	Spe. (%)	Pre. (%)	F1	AUC (ROC)
1DCNN-RLM	84.2	81.7	85.7	77.9	0.798	0.915
1DCNN-RLM + Markov chain	89.5	86.3	91.5	86.3	0.863	0.956
1DCNN-RLM + RNN	89.1	85.1	91.5	86.0	0.856	0.949
1DCNN-RLM + RNN + Markov chain	89.6	85.9	91.9	86.8	0.863	0.950
1DCNN-RLM + BiGRU-TDM	**91.1**	**88.9**	92.4	87.8	**0.883**	**0.970**
1DCNN-RLM + BiGRU-TDM + Markov chain	91.0	87.7	**93.0**	**88.5**	0.881	0.964

Table 9
Comparison of per-segment and per-recording detection results of the proposed method and phased methods on the withheld set. Method 5 is the proposed method. Methods 1–4 are the phased methods previously implemented.

Method	Per-segment detection		Per-recording detection		
	Results (%)	AUC (ROC)	Results (%)	MAE	Corr.
1. 1DCNN-RLM without the 1D-inception block+ BiGRU-TDM without the Self-attention layer	Acc. = 86.1Sen. = 81.7Spe. = 88.8	0.932	Acc. = 97.1Sen. = 95.7Spe. = 100	5.852	0.884
2. 1DCNN-RLM with the original 1D-inception block+ BiGRU-TDM without the Self-attention layer	Acc. = 88.1Sen. = 81.9Spe. = 92.0	0.948	Acc. = 97.1Sen. = 95.7Spe. = 100	4.435	0.946
3. 1DCNN-RLM with the improved 1D-inception block+ BiGRU-TDM without the Self-attention layer	Acc. = 90.5Sen. = 85.1**Spe. = 93.7**	0.963	Acc. = 97.1Sen. = 95.7Spe. = 100	3.798	0.955
4. 1DCNN-RLM with the improved 1D-inception block+ BiGRU-TDM with the original Self-attention layer	Acc. = 90.6Sen. = 87.9Spe. = 92.2	0.963	Acc. = 100Sen. = 100Spe. = 100	3.586	0.962
5. 1DCNN-RLM with the improved 1D-inception block+ BiGRU-TDM with the improved Self-attention layer	**Acc. = 91.1Sen. = 88.9**Spe. = 92.4	**0.970**	**Acc. = 100Sen. = 100Spe. = 100**	**3.333**	**0.968**

4.6. Model optimization

Before determining the proposed method, this study carried out some optimization on 1DCNN-RLM and BiGRU-TDM. As shown in Table 9, we compare the effects of the proposed method (Method 5) with the four previously implemented methods (Methods 1–4). Method 1 is the most primitive method. All blocks in 1DCNN-RLM of this method are replaced by a convolutional layer (kernel size is 3, stride is 1, filter number is 256), and BiGRU-TDM is not employed. Except for the specificity of the segment classification results, the other indexes of the proposed method are better than those of Methods 1–4.

The detection results of Methods1-3 are shown in Table 9. From Method 1 to Method 3, the classification performance is gradually improved. Generally, if the features extracted from the 1DCNN-RLM are more discriminative, it means better classification results can be achieved. Comparing (i), (j), and (k) in Fig. 10, it can be seen that the 1DCNN-RLM with the improved 1D-inception bock has the strongest feature difference. The feature difference of the 1DCNN-RLM without the 1D-inception block is relatively weak. The feature difference of the 1DCNN-RLM with the original 1D-inception block lies between the two. The test results in Table 9 are consistent with this outcome. This indicates that the 1D-inception block can increase the representation learning ability of 1DCNN-RLM. The original Inception block can complete multi-scale feature fusion and realize parameter control [51]. In order to improve the nonlinear expression capability of 1DCNN-RLM while also controlling the overall network parameters of the model, we chose to add Conv1D layers to the 1D-inception block (the branch of MaxPool1D layer only needs to be kept as it is) instead of adding them to the network stem. This improvement also means that 1DCNN-RLM can achieve deeper multi-scale feature fusion and stronger representation learning ability. In this work, there is no strict ablation experiment to verify the effect of modifying the 1D-inception block, but

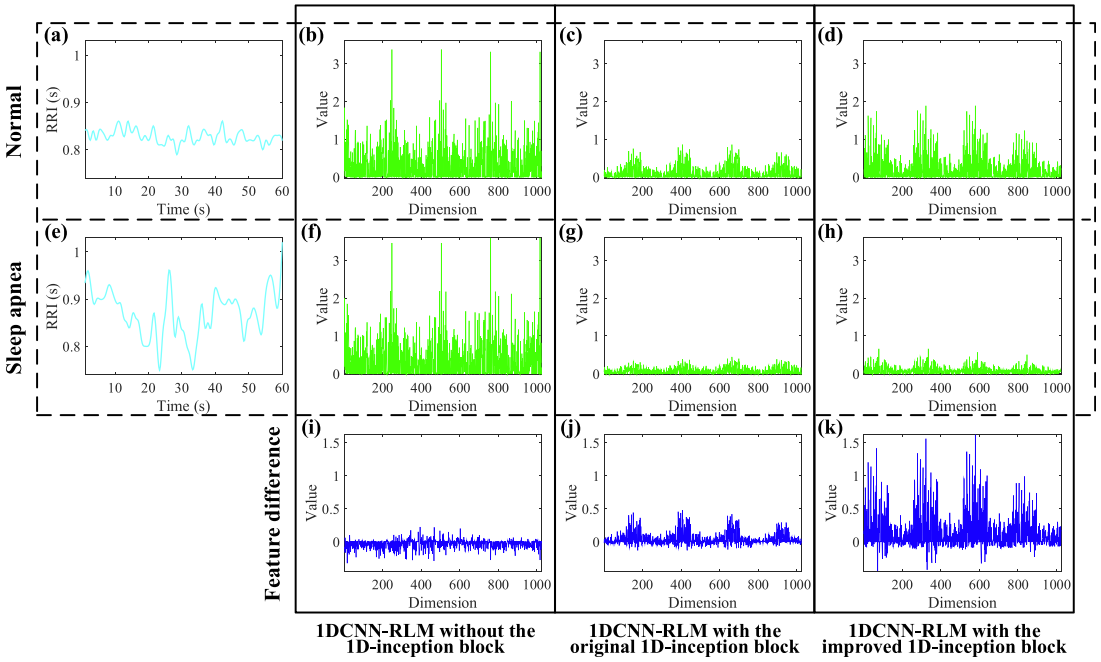


Fig. 10. (a) A normal sample in the withheld set. (b), (c), and (d) are the feature vectors of the normal sample extracted by 1DCNN-RLM without the 1D-inception block, 1DCNN-RLM with the original 1D-inception block, and 1DCNN-RLM with the improved 1D-inception block, respectively. (e) An SA sample in the withheld set. (f), (g), and (h) are the feature vectors of the SA sample extracted by 1DCNN-RLM without the 1D-inception block, 1DCNN-RLM with the original 1D-inception block, and 1DCNN-RLM with the improved 1D-inception block, respectively. (i), (j), and (k) show the difference of feature vectors between the normal sample and the SA sample in 1DCNN-RLM without the 1D-inception block, 1DCNN-RLM with the original 1D-inception block, and 1DCNN-RLM with the improved 1D-inception block, respectively. The feature difference is the vector obtained by subtracting the corresponding dimensions of the normal feature vector and the SA feature vector.

the results of Methods 2–3 in Table 9 are obtained by the method of control variates. The comparison of these results confirms the effectiveness of the modification.

The self-attention mechanism seems to contribute to alleviating the imbalance of sensitivity and specificity according to the results of Methods 3–5 in Table 9. It is remarkable that the methods after introducing self-attention mechanism can completely distinguish normal recordings from SA recordings. We believe that the introduction of long-term temporal dependence is essential to the per-recording detection for better performance. The output probabilities of Method 3, Method 4, and the proposed method are compared in Fig. 9(b). The proposed method uses the improved Self-attention layer. Weights are added between time steps and inside each single time step to highlight important information, which can classify segments near the state transition position more accurately than other two methods. Due to the lack of attention to the internal information of the time step, Method 4 is slightly worse than the proposed method. Method 3 does not use the self-attention mechanism. Although it can correctly predict more normal segments, the classification result is relatively poor. The results in Fig. 9(b) are also consistent with the results in Table 9. The attention mechanism in deep learning draws on human attention behavior. It can use limited attention resources to filter out high-value information from a large amount of information [58]. In this work, each time step in the improved Self-attention layer adds a Softmax layer so that the key information within a single time step is also extracted. The motivation for this improvement is to make this attention mechanism closer to the thinking mode of human beings. An example in NLP: When using a model to parse a sentence, in addition to paying attention to the arrangement of words, the spelling of the word itself should also receive sufficient consideration. Although the role of BiGRU-TDM is mainly to introduce long-term temporal dependence between segments, redundancy should be appropriately dealt with for the information in a single time step. This improvement may reduce the burden of BiGRU-TDM training to some extent, thereby speeding up the convergence of the model. The comparison between Method 4 and Method 5 in Table 9 shows that modifying the Self-attention layer can slightly improve the detection effect, and we believe that the practical significance and necessity of this modification are more prominent.

4.7. Transferability of the proposed method

Before applying the method to practical problems, it is necessary to conduct a generalization experiment to verify the transferability of the method. Especially for deep learning methods, models that lack generalization experimental validation are often considered as overfitting. This is an acute problem because an overfitting model has little practical significance and application prospects. We conceive that two or more databases from different sources should be used for a generalization experiment: one database for model training, and another database from different sources for model testing. This will be close to the actual situation because there may be a substantial time and space span between the attributes of the train set and test set. The generalization experiment results of this study are shown in Table 5, and the AUC (ROC) of the per-segment detection was 0.951. The specificity reached 100% in the per-recording detection, and only an SA patient was misclassified. The MAE and Corr. were 4.762 and 0.934, respectively. The good transferability of this method is a strong proof of its bright application prospects.

4.8. Future research work

The SA detection method based on a single-lead ECG is expected to be inexpensive and convenient in diagnostic process. At present, most patients with sleep apnea may miss the best treatment period due to the lack of timely diagnosis. Hence, we think there is an urgent demand for a diagnostic program that is well-disposed to people. The proposed method takes into account the temporal information of the past and the next twelve minutes, which can be considered as sacrificing the possibility of real-time detection to achieve better detection performance. A wearable detection device development scheme, namely, the single-lead ECG acquisition component of the device completes the overnight signal acquisition, and the embedded software based on the proposed method then performs the detection. Obviously, this is not a real-time strategy, but it has important practical significance for the diagnosis of sleep apnea patients. Finally, it is mentionable that all the methods based on deep learning to date have faced a common problem, that is, the uninterpretability of neural networks. This problem may require continuous exploration and research before it can be resolved.

5. Conclusion

In this study, we propose an SA detection method based on the 1DCNN-RLM and BiGRU-TDM using a single-lead ECG. The 1DCNN-RLM is employed to automatically extract discriminative feature vectors from each RRI series. The 1DCNN-RLM does not rely on the priori knowledge of experts. Taking into account the long-term temporal dependence of SA, this research proposes and constructs the BiGRU-TDM to learn the state (SA/normal) transition pattern between 1-minute ECG segments. BiGRU-TDM can significantly improve classification performance. Furthermore, we utilize the data-level ADASYN sampling method to synthesize the RRI series of the minority class to deal with the imbalanced learning problem. ADASYN can relieve the imbalance between sensitivity and specificity. The experimental results compared with peers confirm the reliability of the proposed method. Furthermore, as an inexpensive and convenient diagnostic solution, this method can be applied to the development of wearable detection equipment. We believe that the proposed method has broad prospects for clinical applications.

CRediT authorship contribution statement

Hengji Qin: Conceptualization, Methodology, Software, Validation, Visualization, Writing – original draft. **Guanzheng Liu:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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